

Remedy

B. Adjoint sensitivities

$$\left. \begin{array}{l} \text{minimize } J(u; p) \\ \text{Subj. to } \frac{du}{dt} = f(u, t, p) \end{array} \right\} \begin{array}{l} \text{(dynamic} \\ \text{equality constrained optimization} \end{array}$$

$$L(u, \lambda; p) = J(u; p) + \int_0^T \lambda^T \left(f - \frac{du}{dt} \right) dt \quad \begin{array}{l} \text{Comp's} \\ \text{Lagrange} \\ \text{multiplier} \end{array} \quad \lambda(t) \in \mathbb{R}^n$$

$$\mathcal{L} = \int_0^T g(u; p) + \lambda^T(t) \left(f - \frac{du}{dt} \right) dt$$

$$\frac{d\mathcal{L}}{dp} = \int_0^T \left[\frac{\partial g}{\partial p} + \frac{\partial g}{\partial u} \frac{du}{dp} + \lambda^T(t) \left(\frac{\partial f}{\partial p} + \frac{\partial f}{\partial u} \frac{du}{dp} - \frac{d}{dt} \frac{du}{dp} \right) \right] dt$$

difficult

Rearrange:

$$\frac{d\mathcal{L}}{dp} = \int_0^T \left[\frac{\partial g}{\partial p} + \lambda^T(t) \frac{\partial f}{\partial p} + \underbrace{\left(\frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} - \lambda^T \frac{d}{dt} \right)}_{\text{make zero}} \frac{du}{dp} \right] dt$$

integrate by parts.

$$\begin{aligned} \int_0^T -\lambda^T \frac{d}{dt} \frac{du}{dp} dt &= \left[-\lambda^T \frac{du}{dp} \right]_0^T + \int_0^T \frac{d\lambda}{dt}^T \frac{du}{dp} dt \\ &= \lambda(0)^T \frac{du}{dp}(0) - \lambda(T)^T \frac{du}{dp}(T) + \int_0^T \frac{d\lambda}{dt}^T \frac{du}{dp} dt \end{aligned}$$

$$\frac{d\mathcal{L}}{dp} = \int_0^T \left[\frac{\partial g}{\partial p} + \lambda^T \frac{\partial f}{\partial p} + \underbrace{\left(\frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} + \frac{d\lambda}{dt}^T \right)}_{\text{make zero}} \frac{du}{dp} \right] dt + \lambda(0)^T \frac{du}{dp}(0) - \lambda(T)^T \frac{du}{dp}(T)$$

Choose

$$\frac{\partial g}{\partial u} + \lambda^T \frac{\partial f}{\partial u} + \frac{d\lambda^T}{dt} = 0 ; \quad \lambda(T) = 0$$

$$\begin{cases} \dot{\lambda} = -f_u^T \lambda - g_u^T \\ \lambda(T) = 0 \end{cases}$$

Terminal value problem,
Solve backwards in time.

$$\frac{dJ}{dp} = \frac{dJ}{dp}$$

Strategy

① Solve forward $\begin{cases} \frac{du}{dt} = f(u, t, p) \\ u(t=0) = u_0 \end{cases}$ for $u(t)$ e.g. RK45

② Solve backward $\begin{cases} \frac{d\lambda}{dt} = -\frac{\partial f}{\partial u}^T \lambda - \frac{\partial g}{\partial u}^T \\ \lambda(t=T) = 0 \end{cases}$ for $\lambda(t)$, e.g. RK45

③ Evaluate: $\frac{dJ}{dp} = \int_0^T \left[\frac{\partial g}{\partial p} + \lambda^T \frac{\partial f}{\partial p} \right] dt + \lambda(0) \frac{du_0}{dp}$ (by quadrature)

- Scales constantly in the number of parameters $p \in \mathbb{R}^m$.